

Insight AI: AI-Powered Data Analytics for Smarter Decisions

Software Requirements and Design Specifications (SRDS)

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1 Introduction

1.1 System Introduction

Insight AI is an AI-powered, web-based data analytics platform designed to democratize data analysis for non-technical users including business managers, educators, and healthcare professionals. The system integrates automated machine learning pipelines with Google Gemini AI to create intelligent analytics assistant capable of transforming raw datasets into actionable insights within minutes. Built on FastAPI and Next.js technologies, the platform enables seamless dataset processing, automated model training, and interactive dashboard generation. Key features include intelligent target column detection, multi-model evaluation with cross-validation, AI-generated narrative insights using natural language generation, automated visualization recommendations, and natural language query processing for conversational data exploration. The system aims to eliminate technical barriers to data analysis, reduce dependency on data scientists, and empower decision-makers with immediate, accurate insights through intuitive dashboards and exportable reports.

1.2 Background of the System

Modern organizations face significant challenges in leveraging data for decision-making, including high dependency on technical specialists due to complex analytics tools, inconsistent data interpretations from manual analysis variability, limited accessibility to advanced machine learning capabilities, and time-consuming preprocessing workflows. Traditional business intelligence platforms like Tableau, Power BI, and SQL-based systems demand extensive technical knowledge, programming skills, and statistical expertise, leading to user frustration and delayed insights. Manual data preprocessing, model selection, and insight generation processes are time-consuming, error-prone, and require specialized training. Organizations struggle to maintain democratized access to data analytics while managing diverse datasets, and there's insufficient automation in end-to-end analytical workflows including data cleaning, feature engineering, model evaluation, and narrative insight generation. The Insight AI platform was conceived to address these challenges by leveraging modern AI technologies including Google Gemini for natural language generation and automated machine learning pipelines to create an intelligent, self-service analytics system that maintains analytical rigor while ensuring accessibility and providing comprehensive, automated insights without requiring technical expertise.

1.3 Objectives of the System

The major objectives of the Insight AI system are:

- Develop a web-based analytics platform that seamlessly handles CSV and Excel dataset uploads with automated processing workflows
- Implement intelligent preprocessing pipelines using Python for comprehensive data cleaning, including missing value imputation, outlier detection, categorical encoding, and automated feature engineering
- Create automated problem detection algorithms that identify target columns and determine whether datasets require classification, regression, or clustering approaches
- Establish multi-model machine learning training capabilities with cross-validation evaluation
- Build AI-powered insight generation using Google Gemini for automated narrative analysis, business recommendations, and data storytelling
- Implement natural language query processing that converts plain English questions into executable SQL and Pandas commands with instant result visualization
- Design interactive dashboard generation with multiple visualization types including bar charts, line graphs, pie charts, scatter plots, and KPI metric cards
- Provide comprehensive analytics output including model performance metrics, feature importance rankings, and exportable PDF reports
- Enable quick, accurate, and accessible data-driven decision-making for non-technical users without requiring programming or statistical expertise

1.4 Significance of the System

Insight AI makes advanced data analytics accessible to everyone by removing technical barriers that have traditionally limited data-driven decision-making. The platform transforms what used to take days of manual work into minutes of automated analysis, helping organizations respond faster to changing conditions. Business managers can now explore sales patterns and customer trends without coding skills, leading to quicker strategic decisions and competitive advantages. Educators gain clear visibility into student performance metrics, enabling targeted interventions that improve learning outcomes. Healthcare administrators can monitor operational efficiency and patient care indicators without relying on technical teams. The system's automated preprocessing and model training ensure consistent, reliable results while eliminating human errors common in manual analysis. Its natural language interface lets users ask questions conversationally and receive instant, understandable answers. By handling all technical complexity automatically, Insight AI frees professionals to focus on what matters most interpreting insights and taking action. This represents a genuine shift in making data analytics a practical tool for everyday decision-makers, not just specialists.

2 Overall Description

2.1 Product Perspective

Insight AI operates between traditional business intelligence tools and modern artificial intelligence technologies. Unlike conventional analytics platforms that require technical expertise or simple visualization tools with limited analytical capabilities, this system integrates deeply with the entire data analysis workflow, from raw data upload to actionable insights. It uses FastAPI for high-performance backend operations, PostgreSQL for reliable data storage, Google Gemini AI for intelligent insight generation, and Next.js for responsive user interfaces. The platform sets itself apart by maintaining the analytical quality of professional tools like Power BI while adding the accessibility and automation of AI-powered analysis. It serves as a bridge between complex statistical methods and everyday business users, enabling organizations to build data-focused cultures without requiring extensive training programs or hiring specialized analysts. The natural language query interface makes data access easier for everyone, allowing users to explore datasets through conversation rather than learning query languages or programming syntax.

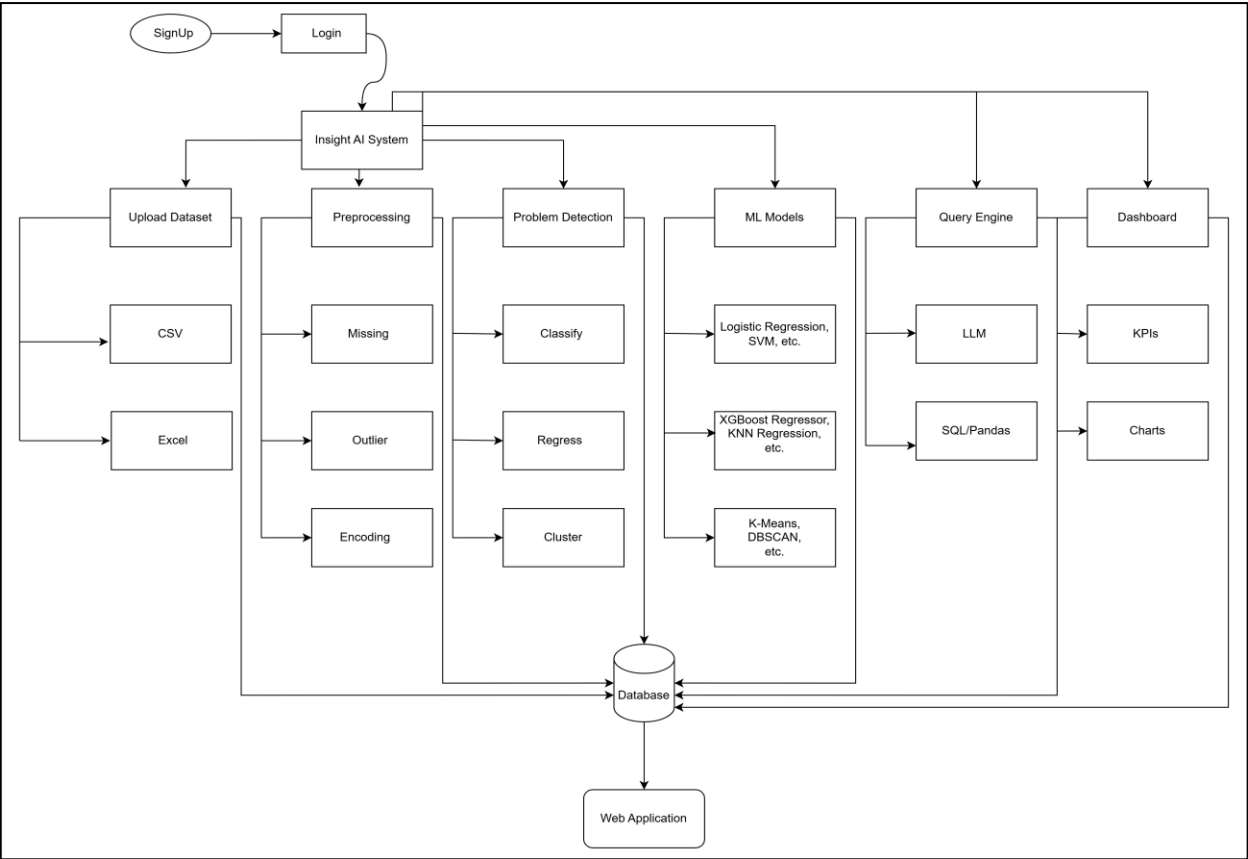


Figure 1: System Design

2.2 Product Scope

The system covers complete management of data analytics workflows, from initial dataset upload through preprocessing to final insight delivery. Its scope includes handling CSV and Excel file uploads through a web interface, automated data cleaning with missing value handling and outlier detection, intelligent detection of target columns and problem types, training multiple machine learning models, generating interactive dashboards with charts and KPIs, creating AI-powered narrative insights using Google Gemini, natural language query processing for conversational data exploration, exporting results as PDF reports, and secure user authentication throughout the analysis process. The system is designed specifically for automated exploratory analysis and predictive modeling, supporting various domains including business analytics, educational assessment, and healthcare monitoring. The platform handles the entire analytical pipeline automatically, enabling users to transform raw data into actionable insights without technical expertise or manual intervention.

2.3 Product Functionality

2.3.1 User Management:

- User registration with email verification and secure password storage
- Login and logout functionality
- Session management for maintaining user access across multiple interactions
- Role-based access control for different user permission levels
- Password recovery and account settings management

2.3.2 Data Upload and Storage:

- Support for CSV and Excel file format uploads through web interface
- File validation to ensure proper format and structure
- Secure storage of uploaded datasets in PostgreSQL database
- Quick preview of uploaded data before processing

2.3.3 Automated Preprocessing Pipeline:

- Missing value detection and automatic filling using median or mode
- Outlier identification and handling through statistical methods
- Categorical variable encoding using label encoding and one-hot encoding
- Feature scaling and normalization for numeric columns
- Automatic removal of ID columns and zero-variance features

2.3.4 Intelligent Problem Detection:

- Automatic target column identification based on data patterns
- Problem type classification (classification, regression, or clustering)
- Feature type categorization (numeric, categorical, datetime)

2.3.5 Machine Learning Engine:

- Training multiple models including Logistic Regression, Random Forest, Decision Trees, etc.
- Cross-validation for reliable performance evaluation
- Automatic model comparison and best model selection
- Feature importance calculation and ranking
- Model performance metrics tracking and storage

2.3.6 AI-Powered Insight Generation:

- Integration with Google Gemini API for natural language insight creation
- Automatic dashboard title generation based on dataset content
- Generation of narrative insights explaining data patterns and trends
- Insight prioritization based on statistical significance
- Business-focused recommendations derived from analytical findings

2.3.7 Natural Language Query Interface:

- Plain English query processing for conversational data exploration
- Automatic conversion of questions into SQL or Pandas commands
- Query execution on stored datasets with result formatting
- Support for aggregations, filtering, and sorting operations
- Real-time query response with visualized results

2.3.8 Visualization and Dashboard:

- Automatic generation of KPI cards showing key metrics
- Multiple chart types including bar charts, line graphs, pie charts, and scatter plots
- Interactive dashboard with persistent visualizations
- Chart recommendations based on data types and relationships
- Export functionality for PDF reports of complete dashboards

2.4 Users and Characteristics

- **Business Manager:**
 - **Description:** Oversees business operations and strategic decision-making using data insights
 - **Activities:** Uploading sales and customer datasets, reviewing generated dashboards, exploring data through natural language queries, exporting reports for presentations, tracking KPIs and business metrics
- **Educator:**
 - **Description:** Uses data to track and improve student performance and learning outcomes
 - **Activities:** Uploading student performance data, analyzing grade distributions, identifying at-risk students through predictive models, generating visualizations for reports, exploring attendance and assessment patterns
- **Healthcare Administrator:**
 - **Description:** Monitors operational efficiency and patient care metrics
 - **Activities:** Uploading patient outcome and operational data, reviewing automated insights on treatment effectiveness, tracking resource utilization, exploring trends through conversational queries, generating compliance reports
- **Data Analyst:**
 - **Description:** Performs quick exploratory analysis and validates initial hypotheses
 - **Activities:** Rapid dataset profiling, automated preprocessing to save time, preliminary model testing, feature importance analysis, generating baseline insights before deeper manual analysis
- **System Administrator:**
 - **Description:** Manages technical infrastructure, user accounts, and system configuration
 - **Activities:** Managing user authentication and access control, monitoring system performance and database health, troubleshooting upload or processing issues, configuring API integrations, maintaining deployed models

2.5 Operating Environment

Insight AI is designed as a web-based application following a modern client-server architecture. The system operates in a distributed environment where the frontend user interface runs on the client's web browser, while the computationally intensive tasks such as data profiling, machine learning model training, and AI insight generation are processed on a centralized backend server. The application relies on a persistent connection to a PostgreSQL database for data storage and requires external network access to communicate with the Google Gemini API for generating natural language insights. The system requires the following hardware, software, and network components to function correctly:

- **Client-Side (User Workstation):**
 - **Device:** Any desktop, laptop, or tablet capable of running a modern web browser.
 - **Operating System:** Platform-independent (Windows, macOS, Linux, Android, or iOS).
 - **Software:** A modern web browser with JavaScript enabled (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari).
 - **Network:** Stable internet connection to upload datasets and view dashboards.
- **Server-Side (Backend Infrastructure):**
 - **Operating System:** Linux (Ubuntu/Debian recommended) or Windows Server.
 - **Runtime Environment:** Python 3.9 or higher (required for FastAPI and Scikit-Learn).
 - **Database Management System:** PostgreSQL 13.0 or higher (specifically required for JSONB support to store datasets).
 - **Hardware Requirements:**
 - **RAM:** Minimum 8GB (16GB+ recommended) to handle in-memory dataframe processing using Pandas.
 - **Storage:** Sufficient SSD storage to house user datasets and trained model metadata.
- **External Dependencies & Interfaces:**
 - **AI Service:** Active internet connection on the server to reach Google Generative AI (Gemini) endpoints.
 - **API Keys:** Valid Google Gemini API credentials configured in the server environment.

3 Specific Requirements

3.1 Functional Requirements

3.1.1 User Account Management

Requirement Number	Requirement	Description
FR-001	User Registration	Users must be able to create a new account using email, social media login, or other supported authentication methods.
FR-002	User Login	Users shall log in with their registered credentials or through single sign-on (SSO) methods.
FR-003	Account Recovery	In case of forgotten credentials, users should be able to recover their accounts through email verification or other secure methods.
FR-004	Profile Management	Users should be able to update their profile information, including profile picture, bio, and personal preferences.

3.1.2 Dataset Preprocessing Pipeline

Requirement Number	Requirement	Description
FR-005	Dataset Upload	The system shall allow users to upload CSV and Excel datasets.
FR-006	Missing Value Handling	The system shall automatically detect and handle missing values (e.g., by replacing with mean or median).
FR-007	Outlier Detection	The system shall detect and manage outliers in the dataset.
FR-008	Categorical Encoding	The system shall perform encoding of categorical variables for ML processing.

3.1.3 Target Column & Model Selection

Requirement Number	Requirement	Description
FR-009	Problem Type Detection	The system shall analyze dataset columns to determine whether the task is classification, regression, or clustering.
FR-010	Classification Selection	If the target column contains categorical labels (e.g., Yes/No), the system shall choose classification models.
FR-011	Regression Selection	If the target column contains continuous numeric values, the system shall choose regression models.
FR-012	Clustering Selection	If no suitable target column exists, the system shall select unsupervised learning models (e.g., K-Means).

3.1.4 Natural Language Query Engine

Requirement Number	Requirement	Description
FR-013	Natural Language Input	The system shall provide a field for users to enter queries in plain English.
FR-014	Query Preprocessing	The system shall preprocess the query (tokenization, stop-word removal, etc.).
FR-015	LLM Query Translation	The system shall use an LLM (e.g., GPT or Gemini) to translate the natural language query into SQL/Pandas code.
FR-016	Query Execution	The system shall execute the translated SQL/Pandas command and return results.

3.1.5 Machine Learning Module

Requirement Number	Requirement	Description
FR-017	Supervised Models	The system shall apply supervised models such as Logistic Regression, Decision Tree, Random Forest, Linear Regression, etc.
FR-018	Unsupervised Models	The system shall apply unsupervised models such as K-Means clustering when applicable.

3.1.6 AI-Powered Insights

Requirement Number	Requirement	Description
FR-019	Automated Insight Generation	The system shall use Google Gemini AI to automatically generate business insights from data profiles, correlations, and ML results.
FR-020	Insight Categorization	Each insight shall be categorized by type (data quality, patterns, statistical, business) and assigned a priority level (high, medium, low).
FR-021	Actionable Recommendations	Every insight shall include a concrete, actionable recommendation section to guide decision-making.

3.1.7 Visualization & Dashboard Module

Requirement Number	Requirement	Description
FR-022	Automatic Chart Generation	The system shall generate suitable visualizations such as bar, pie, line, and scatter charts.
FR-023	KPI Display	The system shall display KPIs, predictions, and insights on the dashboard.
FR-024	Interactive Dashboard	The system shall provide an interactive dashboard allowing users to view multiple insights side-by-side.

3.2 Behaviour Requirement

3.2.1 Use Case Diagram for Insight AI

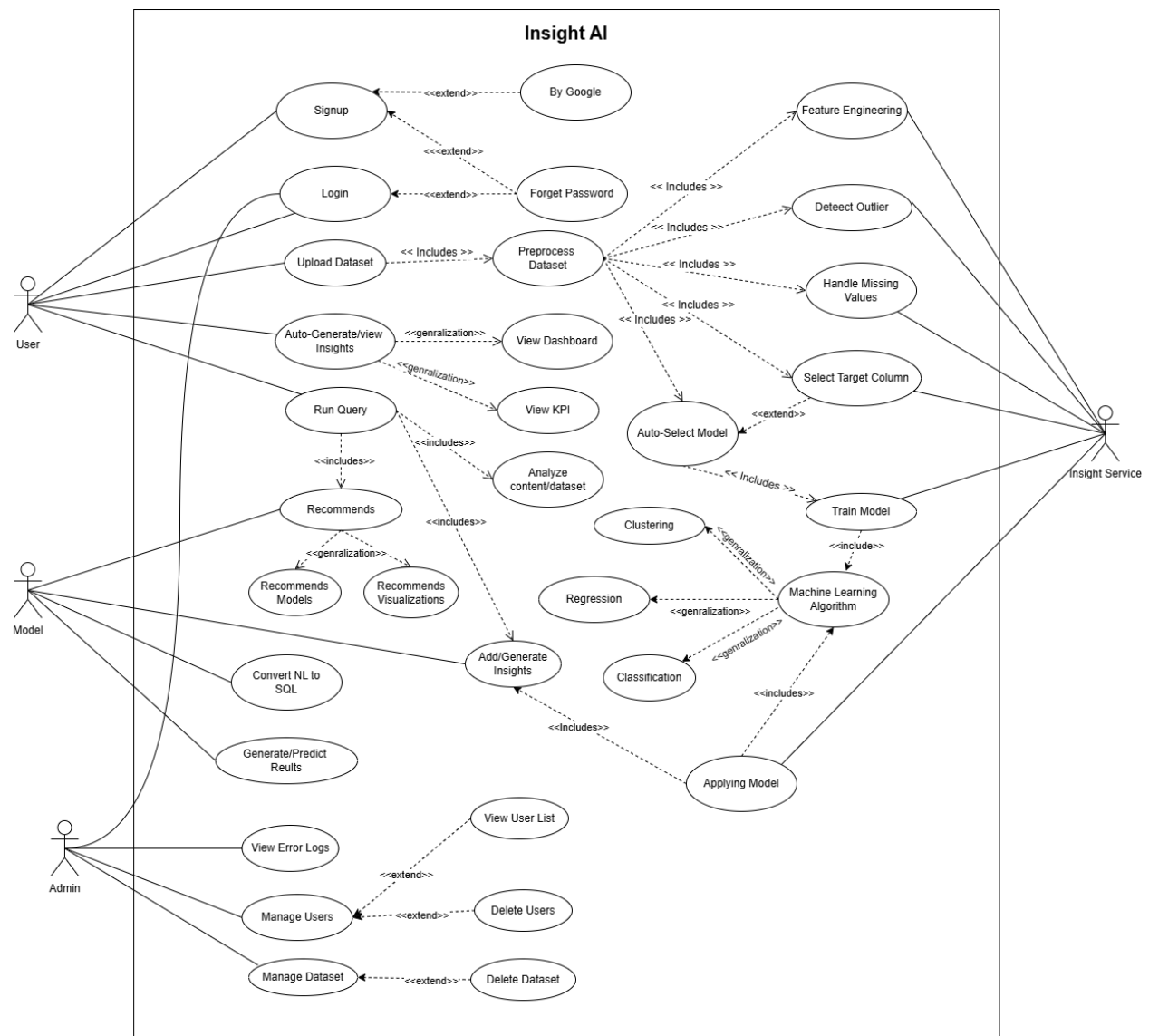


Figure 2: Use Case Diagram of System

3.2.2 Use Case #1: Signup

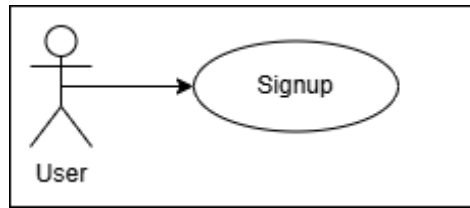


Figure 3: Use Case Diagram for Signup Events

Actors	User
Description	User needs to signup if they don't already have an account.
Pre-Condition	User must have an Email account.
Steps	1. Enter Name, Email and other required information. 2. Press "Signup" after filling relevant details.
Post-Condition	User will be successfully registered with the system.

Table 1: For Signup Events

3.2.3 Use Case #2: Login

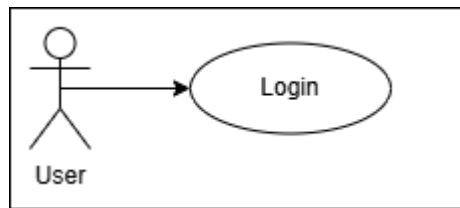


Figure 4: Use Case Diagram for Login Events

Actors	User
Description	User needs to Login to access the features of the system to the fullest.
Pre-Condition	User wants to enter system
Steps	1. Enter Email and Password. 2. Press "Login" after filling relevant credentials.
Post-Condition	Successfully logged in to the system.

Table 2: For Login Events

3.2.4 Use Case #3: Logout

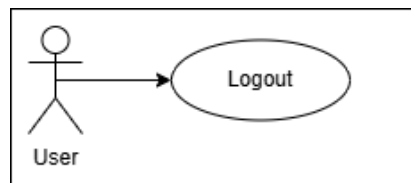


Figure 5: Use Case Diagram for Logout Events

Actors	User
Description	User wants to logout after using the system.
Pre-Condition	User must be logged in to the system.
Steps	1. Navigate to profile section. 2. Press the “Logout” button.
Post-Condition	Successfully logged out of the system.

Table 3: For Logout Events

3.2.5 Use Case #4: Upload data

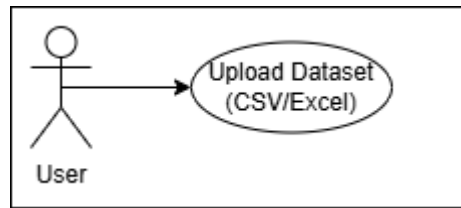


Figure 6: Use Case Diagram for Upload data Events

Actors	User
Description	User wants to upload a dataset (CSV/Excel) to the system.
Pre-Condition	User must be logged in to the system.
Steps	1. Navigate to the dataset upload section. 2. Select a CSV or Excel file from the device. 3. Click the “Upload” button to submit the dataset.
Post-Condition	Dataset is successfully uploaded and ready for processing.

Table 4: For Upload data Events

3.2.6 Use Case #5: Recommend Models

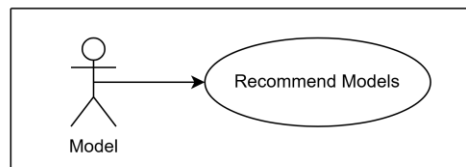


Figure 7: Use Case Diagram for Recommend Models Events

Actors	Model
Description	The system’s model component recommends suitable machine-learning algorithms based on the dataset type.
Pre-Condition	Dataset must be uploaded and preprocessed.
Steps	1. Analyse dataset structure. 2. Identify the problem type (classification, regression, clustering). 3. Recommend appropriate algorithms.
Post-Condition	User receives a list of recommended models.

Table 5: For Recommend Models Events

3.2.7 Use Case #6: Generate Predictions

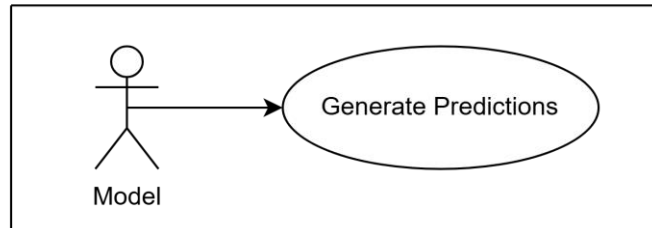


Figure 8: Use Case Diagram for Generate Predictions Events

Actors	Model
Description	Model generates predictions using a trained machine-learning model.
Pre-Condition	A model must be trained and validated.
Steps	1. Load trained model. 2. Apply model to the dataset. 3. Generate predictions.
Post-Condition	Predictions are produced and displayed to the user.

Table 6: For Generate Predictions Events

3.2.8 Use Case #7: Manage Users

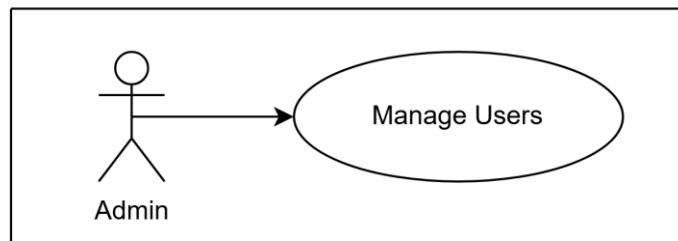


Figure 9: Use Case Diagram for Manage Users Events

Actors	Admin
Description	Admin manages user accounts within the system. Suitable machine-learning algorithms based on the dataset type.
Pre-Condition	Admin must be logged in with administrator access.
Steps	1. Open user management panel. 2. View list of registered users. 3. Add, edit, or remove user accounts.
Post-Condition	User records are updated successfully.

Table 7: For Manage Users Events

3.2.9 Use Case #8: Manage Datasets

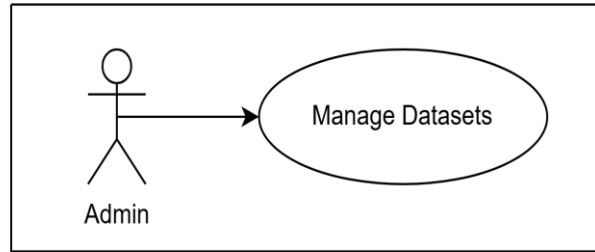


Figure 10: Use Case Diagram for Manage Datasets Events

Actors	Admin
Description	Admin oversees the datasets uploaded by users.
Pre-Condition	Admin must be logged in.
Steps	1. Navigate to dataset management section. 2. View all uploaded datasets. 3. Delete or review specific datasets if required.
Post-Condition	Dataset list is updated as per admin actions.

Table 8: For Manage Datasets Events

3.2.10 Use Case #9: Train Model

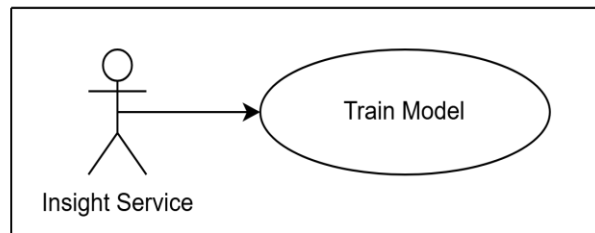


Figure 11: Use Case Diagram for Train Model Events

Actors	Insight Service
Description	The system automatically trains a model using selected algorithms.
Pre-Condition	Dataset must be processed, and model type must be selected.
Steps	1. Receive selected machine-learning model. 2. Split data into training and testing sets. 3. Train the model on training data.
Post-Condition	Trained model is saved and ready for evaluation.

Table 9: For Train Model Events

3.2.11 Use Case #10: Provide Insights

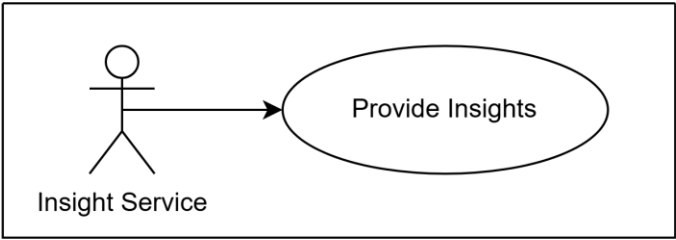


Figure 12: Use Case Diagram to Provide Insights

Actors	Insight Service
Description	The system generates insights such as visualizations, KPIs, and statistical summaries.
Pre-Condition	Dataset must be uploaded and processed.
Steps	1. Analyze dataset. 2. Identify patterns, correlations, KPIs. 3. Generate visualizations and insight summaries.
Post-Condition	Insights are displayed to the user in the dashboard.

Table 10: For Provide Insights Events

3.3 External Interface Requirements

3.3.1 User Interfaces

The Insight AI user interface will be a clean, modern, and intuitive web application. The design will prioritize ease of use for users.

Authentication Interfaces:

- **Login Screen:** Email/password form with validation and error display
- **Registration Screen:** Signup form with email, password strength indicator, and name fields
- **Password Recovery:** Email-based password reset interface

Dashboard Management Interfaces:

- **Create Dashboard:** File upload interface (CSV/Excel) with drag-and-drop support, title/description fields, and target column selection
- **Dashboard Detail View:** Comprehensive view showing dataset statistics, visualizations, ML results, and AI insights.

Data Visualization Interfaces:

- **Auto-Generated Charts:** Grid layout with bar charts, pie charts, histograms, and scatter plots based on data type.

Machine Learning Interfaces:

- **Model Results Panel:** Table displaying trained models with test scores, CV scores, and training times.

AI Insights Interface:

- **Insights Feed:** Card layout with priority badges, titles, and expandable content
- **Insight Categories:** Tabs for Data Quality, Patterns, Statistical, and Business insights

3.3.2 Hardware Interfaces

There is no direct hardware interfaces required for this software, as it is a web-based application. The system operates within the standard hardware of the user's computer (for the browser) and the host server (for the backend)

3.3.3 Software Interfaces

- **Operating Systems:** The client-side (frontend) will interface with any modern OS running a web browser (Windows, macOS, Linux). The server-side will be deployed on a Linux or Windows server environment.
- **Database:** The system will interface with a PostgreSQL database for data storage.
- **Python Libraries:** The backend will interface with Python data science libraries (e.g., Pandas for data manipulation, Scikit-learn for ML models).
- **APIs:** The system will interface with an external LLM API (e.g., GPT or Gemini) to translate natural language queries.

3.3.4 Communications Interfaces

- **HTTP/HTTPS:** Communication between the Next.js client (browser) and the Fast API backend will be via HTTP/HTTPS RESTful API calls.
- **Database Connection:** The backend will use standard database connection protocols to communicate with the PostgreSQL database.
- **External API:** Communication with the external LLM will be via secure HTTPS API calls.

4 Other Non-functional Requirements

4.1 Performance Requirements

- **Dataset Upload Processing:** File upload and initial validation must complete within 3 seconds for files up to 10 MB.
- **Query Response:** Natural language queries resulting in simple data retrieval should be executed and visualized in under 5-7 seconds.
- **Insight Generation:** AI-powered insight creation using Google Gemini API must complete within 10 seconds
- **Model Training:** Initial model analysis upon dataset upload should be completed within 60 seconds for a medium-sized dataset.
- **Chart Data Loading:** Individual chart data retrieval and rendering must be completed within 10 seconds

4.2 Safety and Security Requirements

- **Authentication:** All access to data analysis must be protected by user authentication (login/password).
- **Authorization:** Users must only be able to see, query, and analyze datasets that they have personally uploaded. Data must be segregated by user.
- **Data Privacy:** Uploaded datasets and generated insights must be stored securely. All communication (especially data upload and API calls) must be used by HTTPS.
- **Query Sanitization:** The Natural Language Query Engine must sanitize all generated SQL/Pandas commands to prevent injection attacks.

4.3 Software Quality Attributes

- **Usability:** The system must be highly usable by its target non-technical audience. The interface should be intuitive, requiring no formal training.
- **Maintainability:** The backend logic will be encapsulated in a Fast API, and preprocessing logic will be built into a discrete Python pipeline, making them easy to update and maintain separately from the front end.
- **Flexibility/Adaptability:** The system is designed to handle any CSV or Excel dataset, not just a specific domain. The model selection module is designed to adapt the analysis to the data provided.

5 Design Description

5.1 Composite Viewpoint

5.1.1 Package Diagram (Logical)

This diagram likely visualizes the physical and logical deployment of the application, showing how the client-side interface connects to the server-side infrastructure. It demonstrates the containerization or organization of the Next.js frontend, the Fast API backend, and the database, highlighting how these distinct components interface to form a cohesive web application.

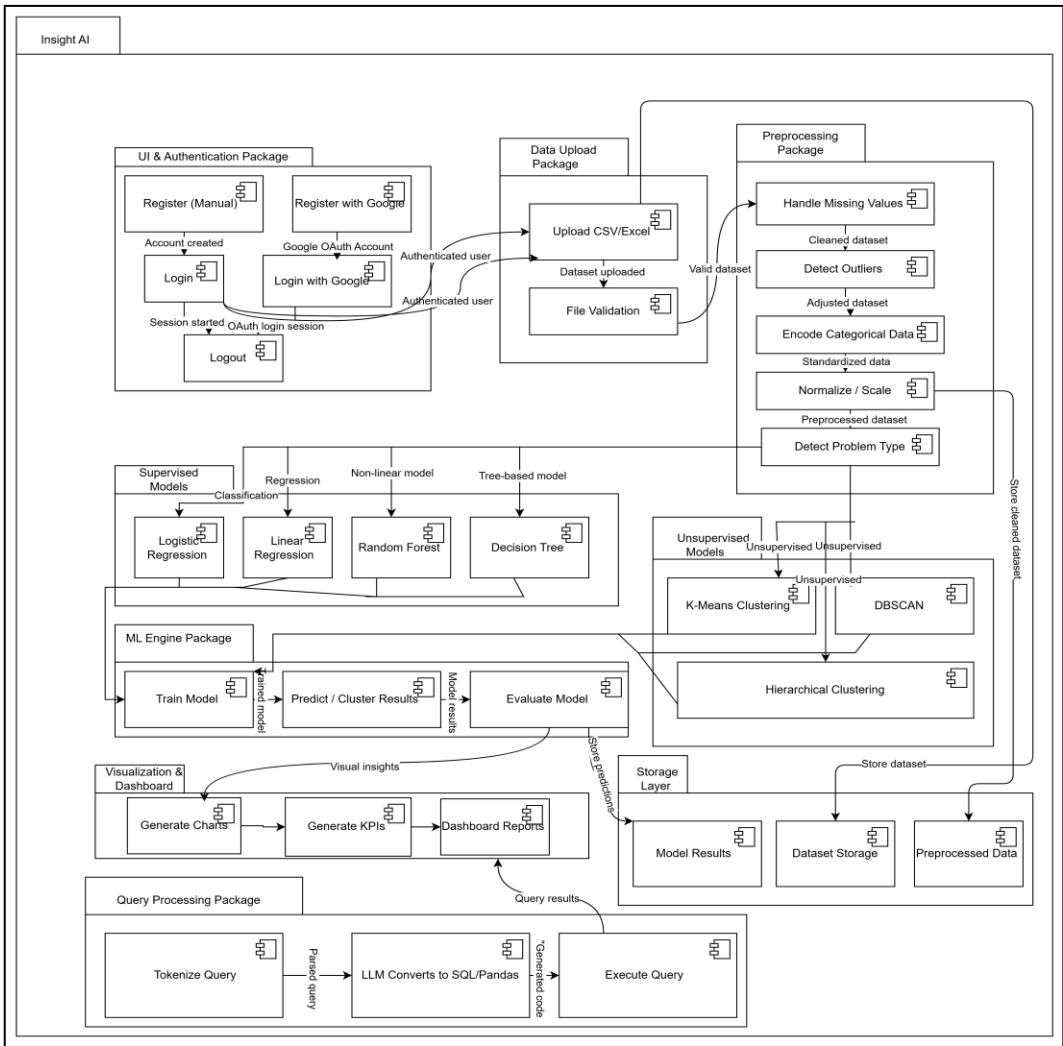


Figure 13: Package Diagram of the System

5.2 Logical Viewpoint

This diagram breaks down the internal software architecture into distinct logical layers or modules, separating the User Interface from the Business Logic (Preprocessing, Model Selection) and Data Access layers. It highlights the modular design where specific components, like the Natural Language Engine or the ML Module, operate independently but communicate to process user request

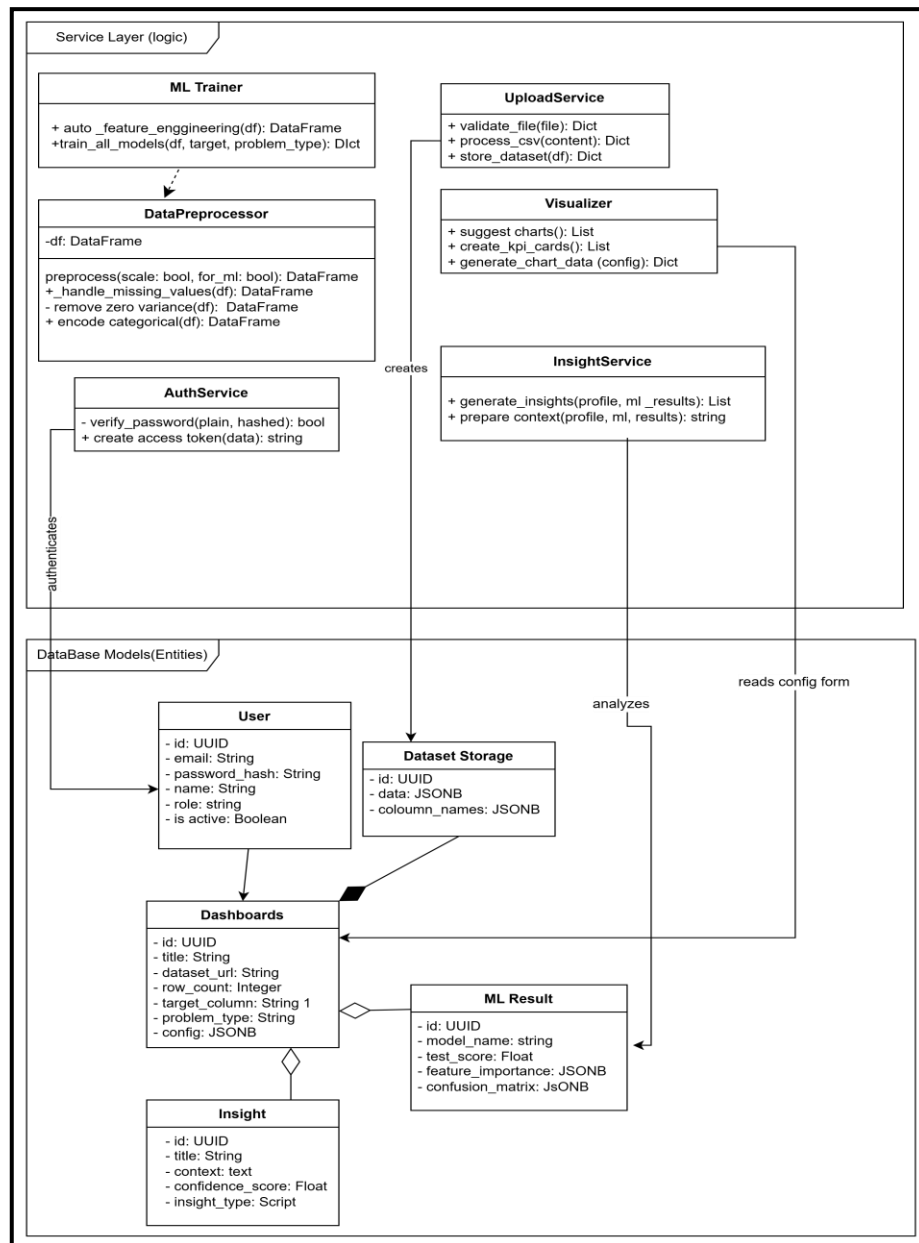


Figure 14: Class Diagram of the System

5.3 Information Viewpoint

5.3.1 LOGIN Sequence Diagram

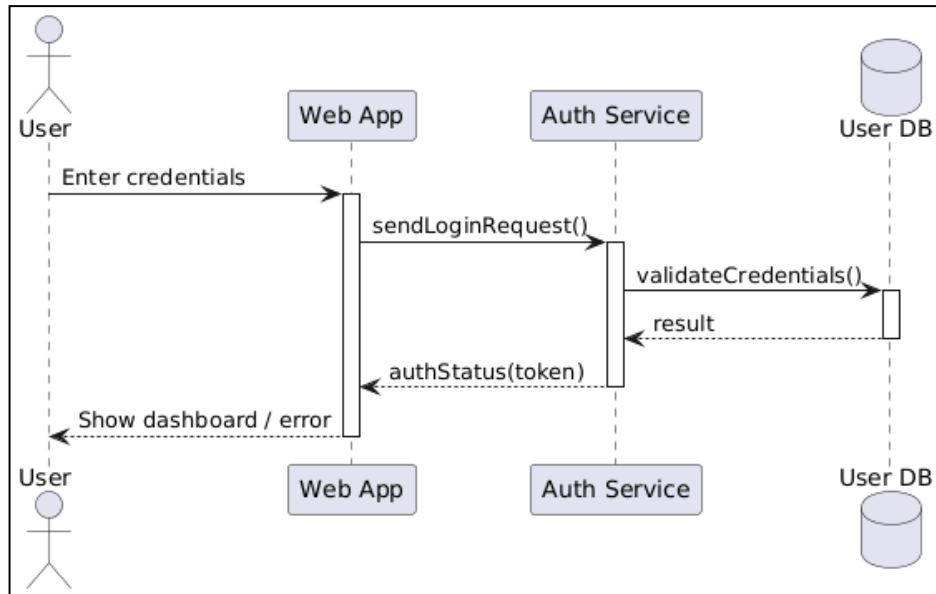


Figure 15: Sequence Diagram for LOGIN

5.3.2 Signup Sequence Diagram

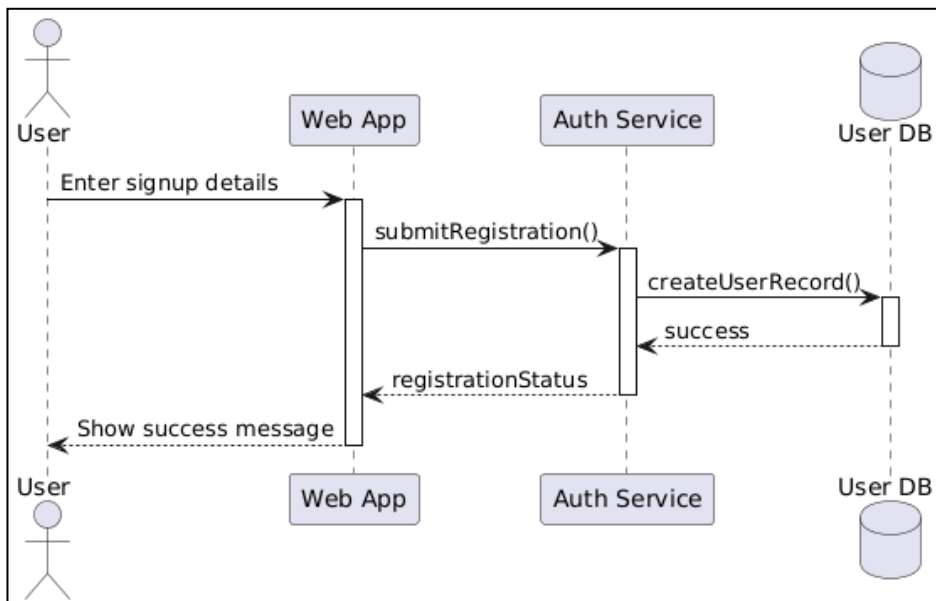


Figure 16: Sequence Diagram for LOGIN

5.3.3 DATASET UPLOAD Sequence Diagram

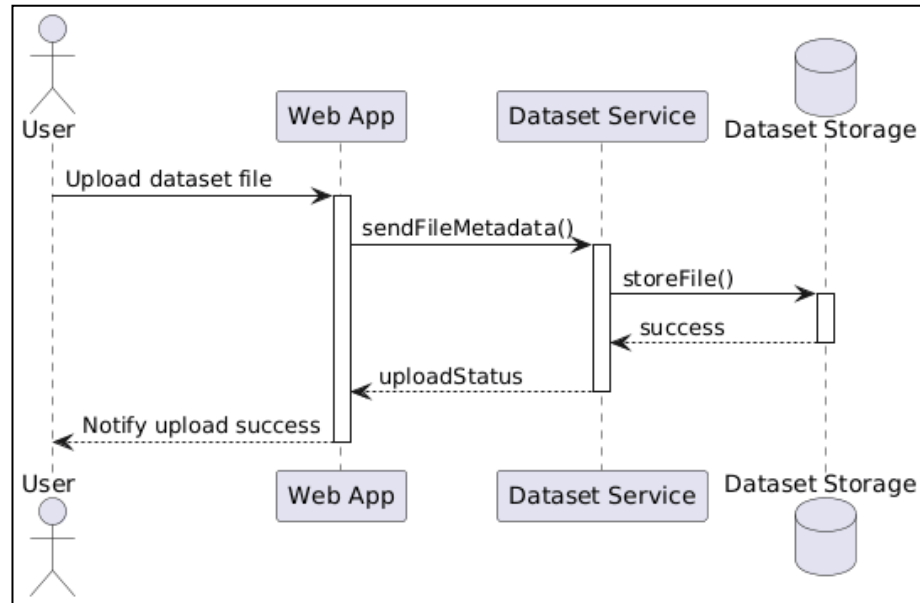


Figure 17: Sequence Diagram for DATASET UPLOAD

5.3.4 Generate Insights Sequence Diagram

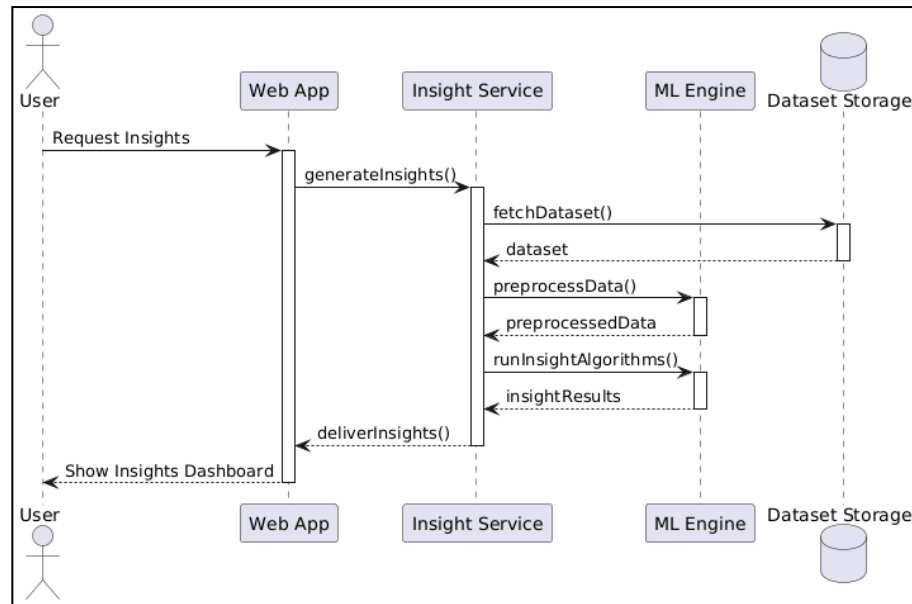


Figure 18: Sequence Diagram for Generate Insights

5.3.5 Run Query Sequence Diagram

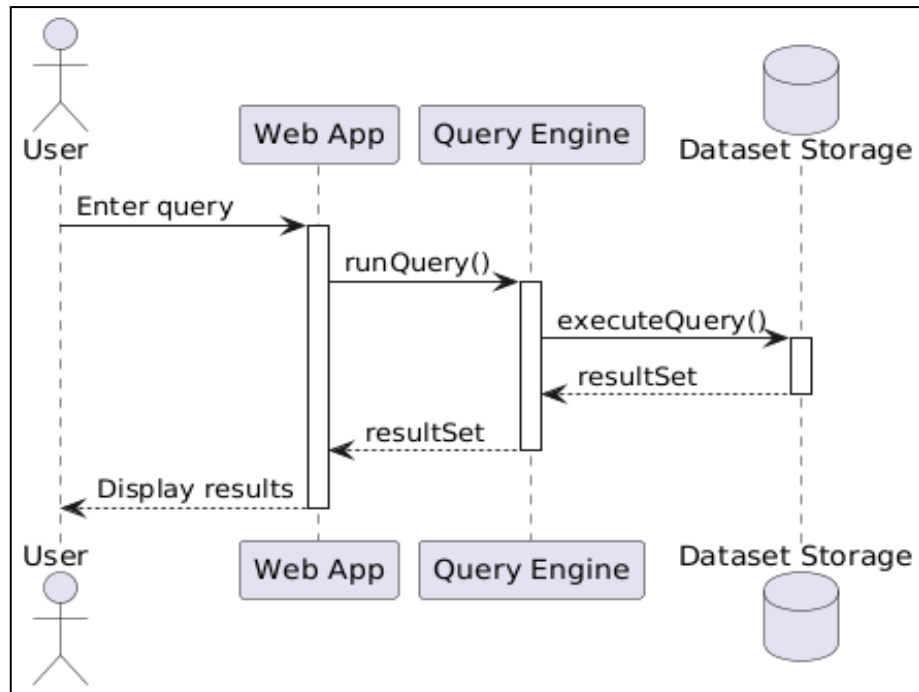


Figure 19: Sequence Diagram for Run Query

5.3.6 Train Model Sequence Diagram

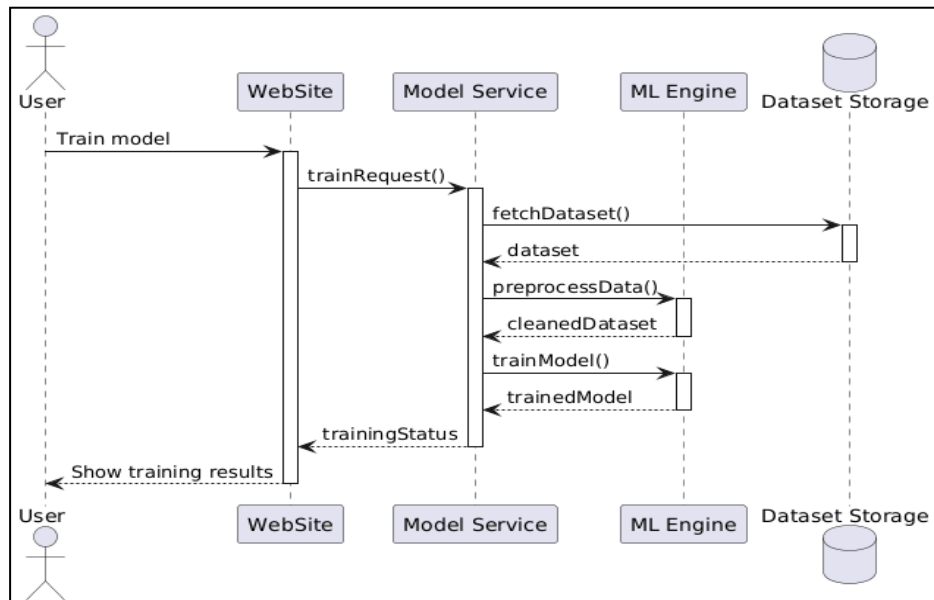


Figure 20: Sequence Diagram for Train Model

5.3.7 Preprocess Data Sequence Diagram

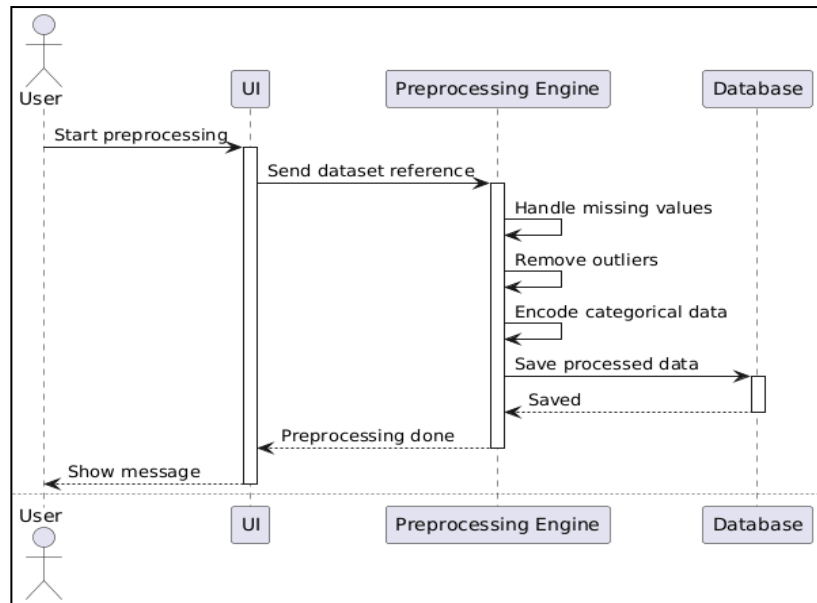


Figure 21: Sequence Diagram for Preprocess Data

5.3.8 View KPI Sequence Diagram

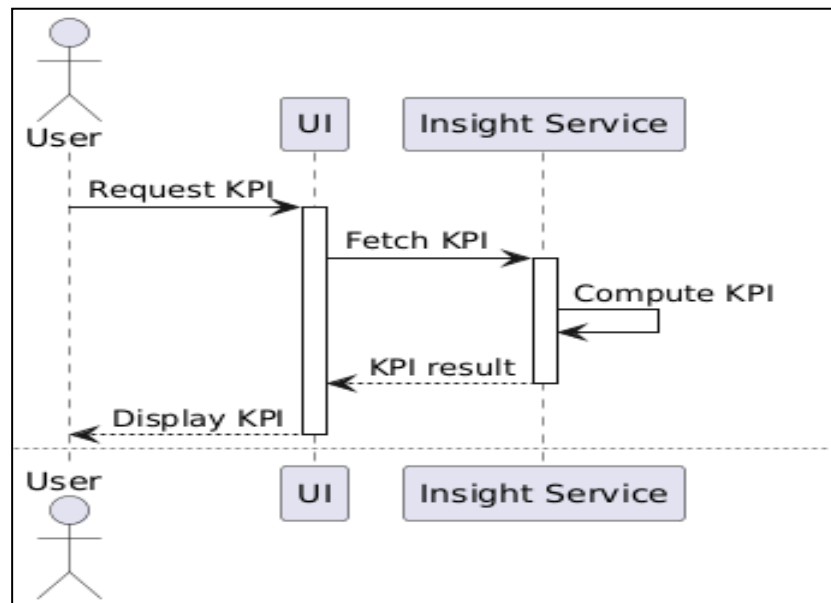


Figure 22: Sequence Diagram for View KPI

5.3.9 View Dashboard Sequence Diagram

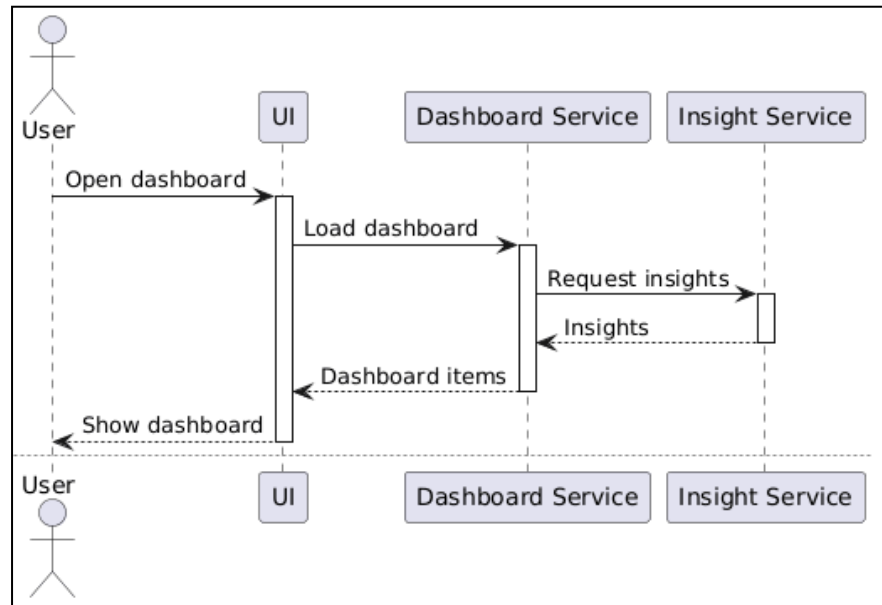


Figure 23: Sequence Diagram for View Dashboard

5.3.10 Recommendation Sequence Diagram

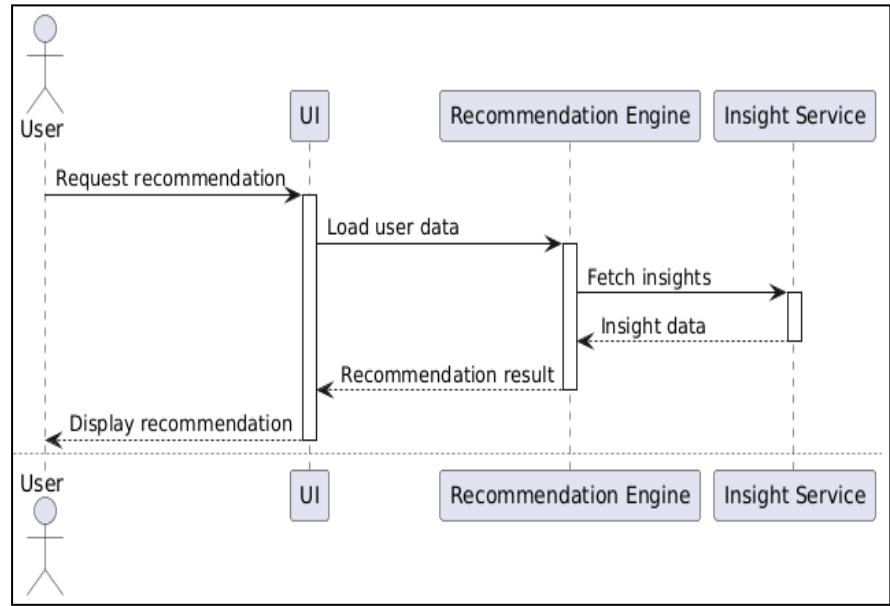


Figure 24: Sequence Diagram for Recommendation

5.3.11 Generate Prediction Sequence Diagram

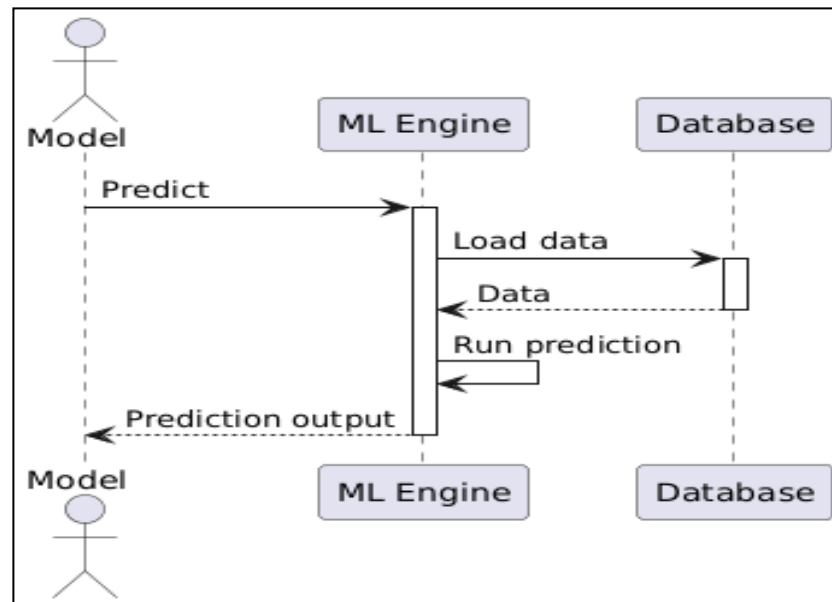


Figure 25: Sequence Diagram for Generate Prediction

5.3.12 Activate Model Sequence Diagram

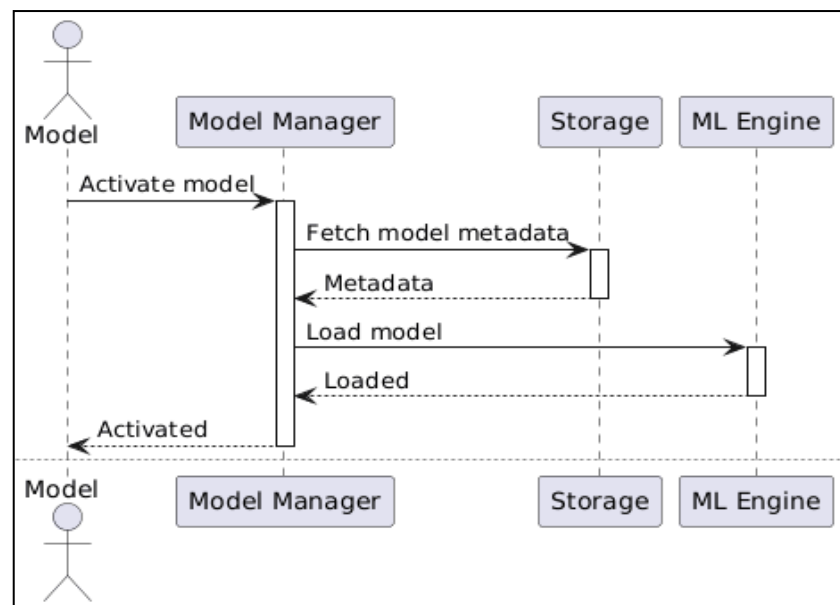


Figure 26: Sequence Diagram for Activate Model

5.3.13 Detect Outliers Sequence Diagram

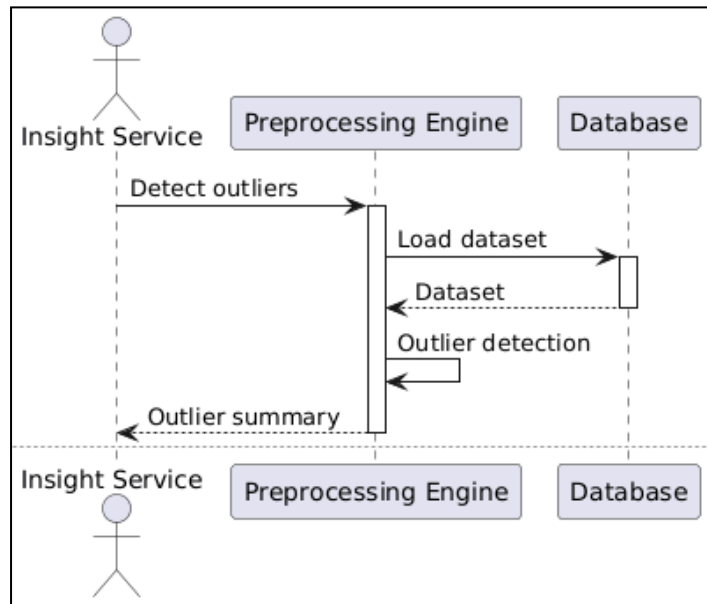


Figure 27: Sequence Diagram for Detect Outliers

5.3.14 Feature Engineering Sequence Diagram

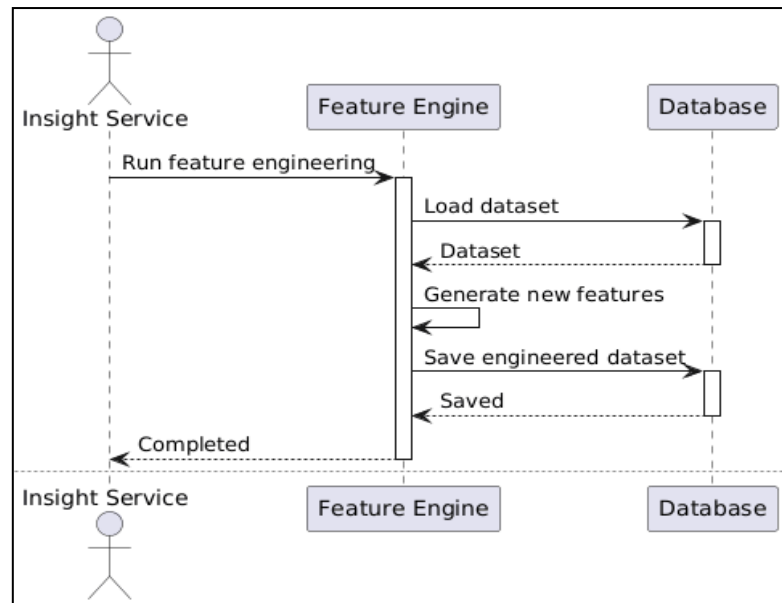


Figure 28: Sequence Diagram for Feature Engineering

5.3.15 Manage Users Sequence Diagram

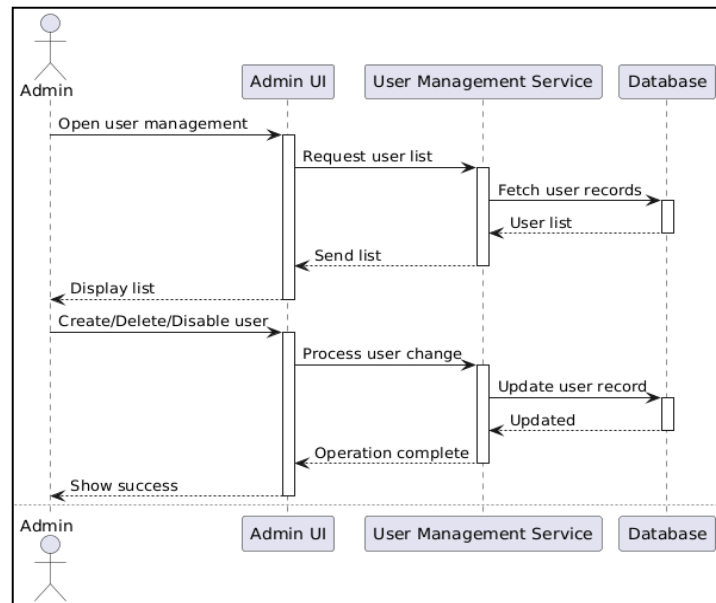


Figure 29: Sequence Diagram for Generate Insights

5.3.16 Manage Users Sequence Diagram

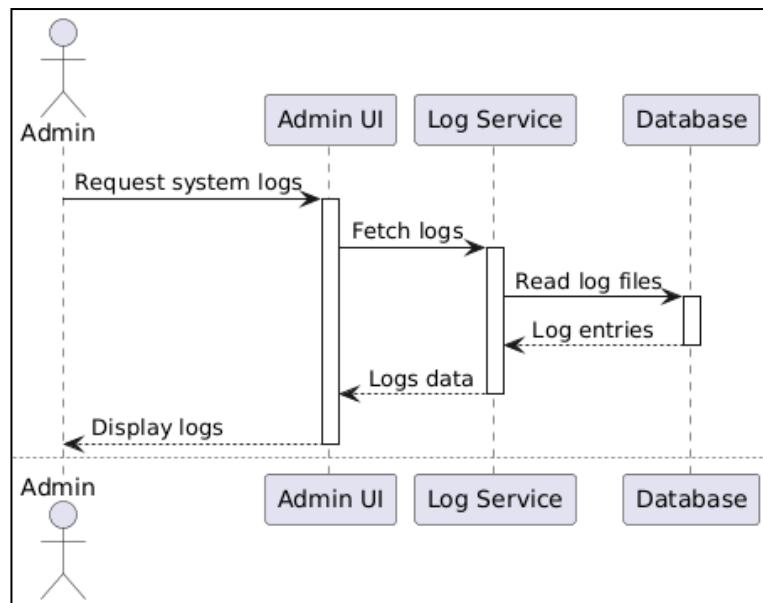


Figure 30: Sequence Diagram for Manage Users

5.4 State Dynamics Viewpoint

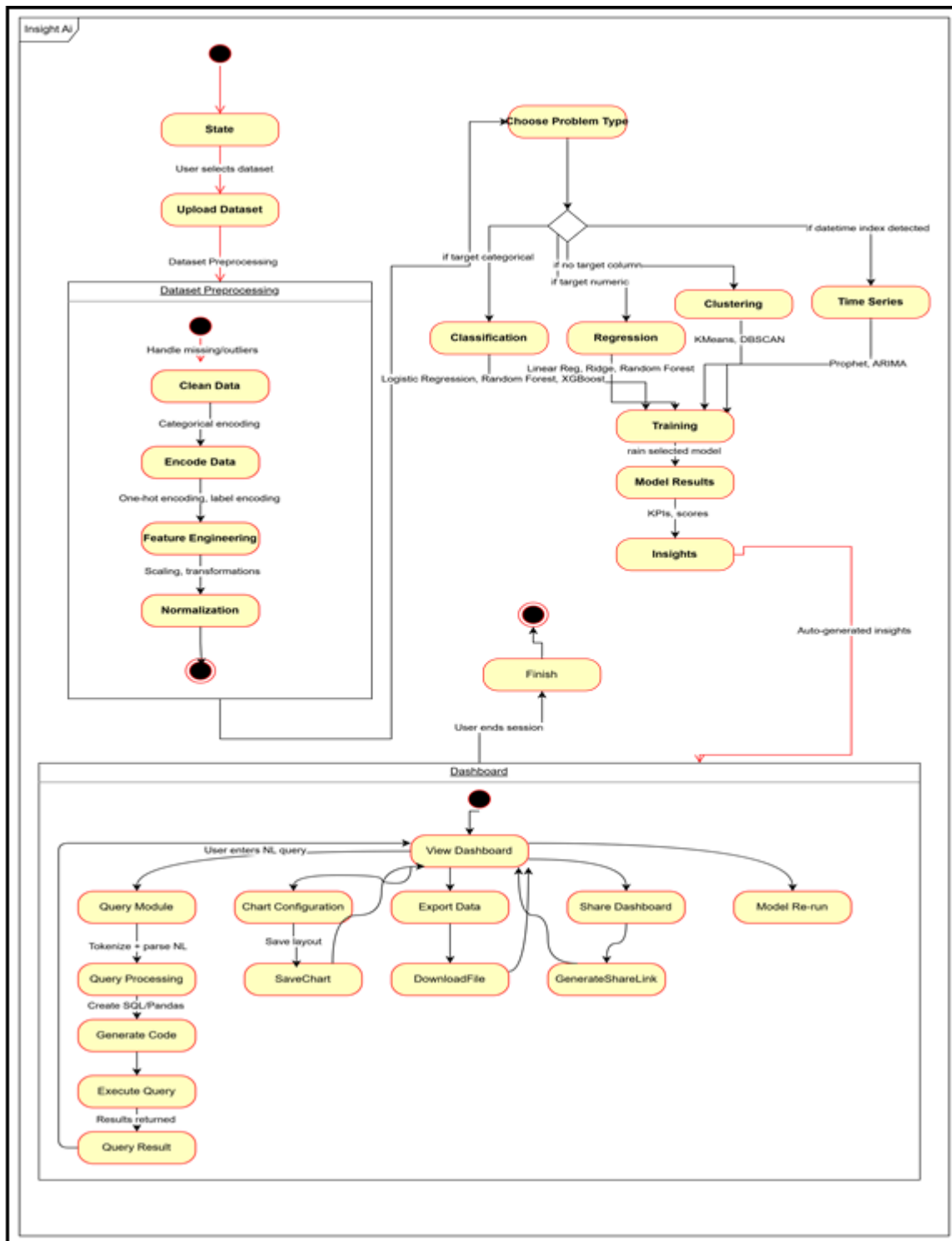


Figure 31: State Diagram for Generate Insights

5.5 Algorithm Viewpoint

This diagram depicts the lifecycle and status transitions of a dataset within the system, such as moving from "Uploaded" to "Preprocessed," then to "Analyzing," and finally "Visualized." It ensures the system handles data reliability by defining that a model cannot be trained until the data has successfully passed the cleaning and outlier detection states.

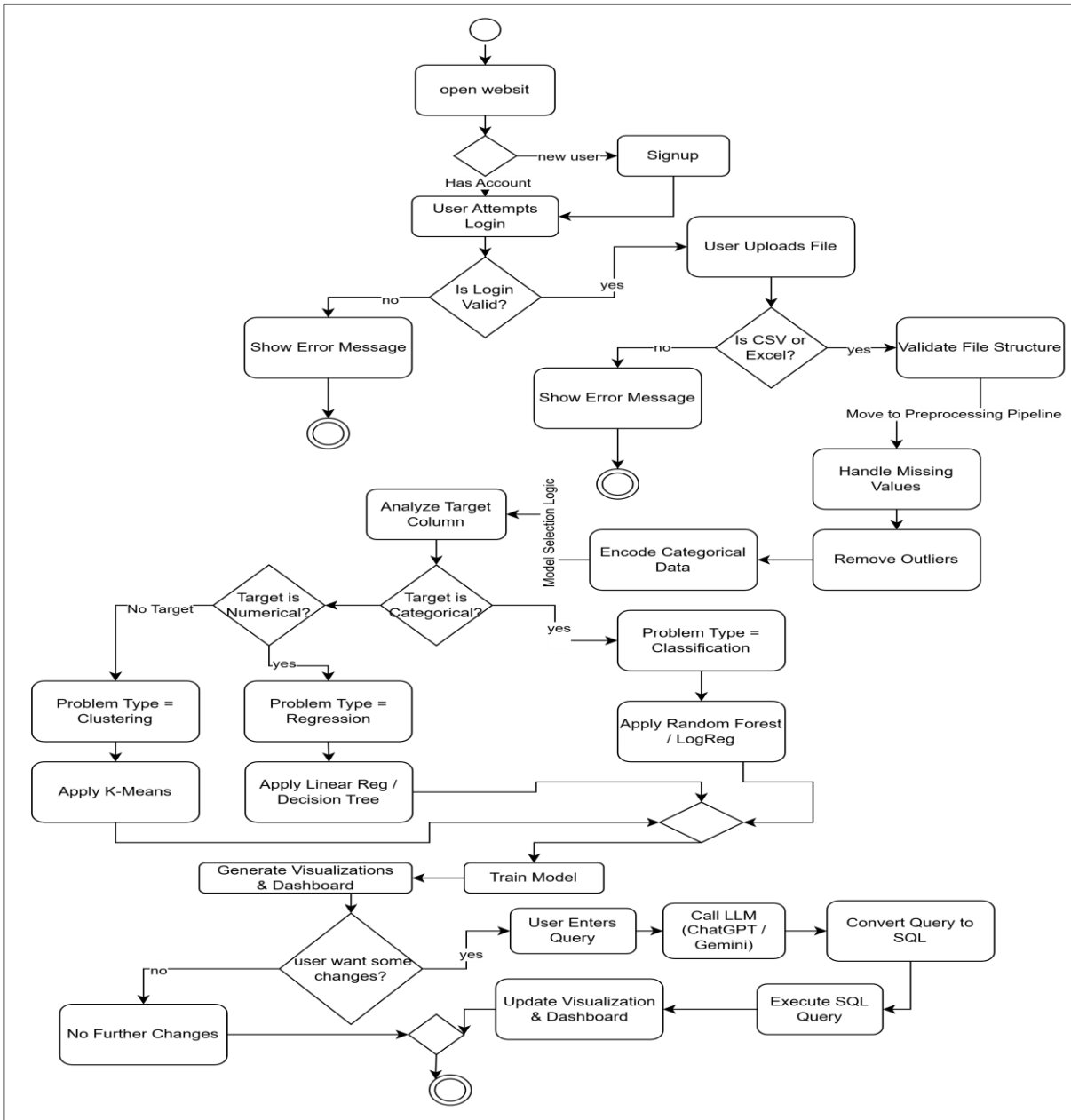


Figure 32: Activity Diagram of the System

5.6 Information Viewpoint

This diagram visualizes the decision-making logic the system uses to automatically select the correct Machine Learning model based on the data structure. It flows through checks on the "Target Column" branching to Classification for categorical data, Regression for numeric values, or K-Means Clustering if no target is provided.

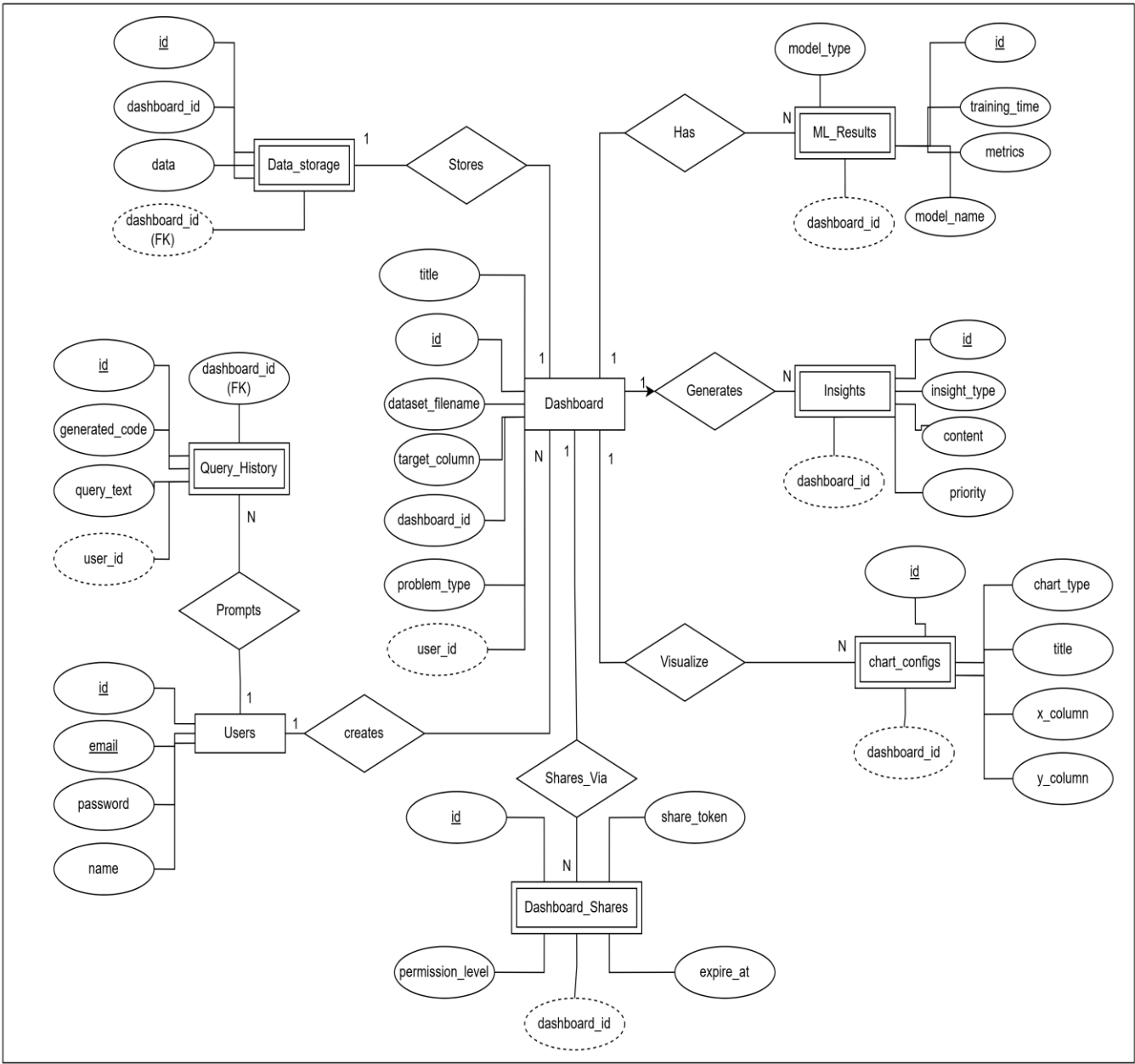


Figure 33: Entity Relationship Diagram of the System